

NFC GR Branch  
L3 Release Candidate  
Equivalence Certificate  
Schema v0.34.0

Generated by NFC Governance Engine v34

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**Document purpose.** This certificate is the formal L3 supersession review document for the NFC Gravity (GR) branch-successor pilot. It maps every generated machine ID to its canonical source, verifies the five-stage closure chain, documents the boundary conditions of domain-bounded CERT-CLOSE, and records the L3 governance decisions required before any supersession of the canonical GR Branch Book may be considered.

## GR Branch L3 Equivalence Certificate

**NFC Governance Engine — GR Branch**  
L3 Release Candidate Equivalence Certificate  
Schema v0.34.0

**Computed branch posture:** domain-bounded-cert-close

**L3 supersession gate:** PASSED (9/9 criteria)

**Active frontier:** ob:GR.global-subcriticality (global curvature-subcriticality — not superseded)

**Supersession effect:** none; this is an L3 release candidate, not a supersession of the canonical GR Branch Book

*This certificate maps each generated machine ID to its canonical source in the NFC\_GR\_Branch book, verifies the proof chain, and records the L3 governance decisions. A domain expert reviewing this certificate can verify that: (i) the computed posture follows from the stated dependencies; (ii) all open frontiers are preserved; (iii) claim force does not exceed canonical force; (iv) no hidden supplementation has occurred.*

## L3 Supersession Criteria

| Criterion                                                                           | Status | Evidence                                                     |
|-------------------------------------------------------------------------------------|--------|--------------------------------------------------------------|
| Every claim has a stable machine ID                                                 | ✓      | All 15 GR claims have stable machine IDs                     |
| Claim force is equivalent to canonical, or changed with explicit release-diff basis | ✓      | Computed posture = domain-bounded-cert-close; canonical post |
| Branch posture reconciles with current release snapshot                             | ✓      | Equivalence report status = passed                           |
| Open frontiers are preserved and correctly labeled                                  | ✓      | ob:GR.global-subcriticality preserved as active frontier = T |
| Theorem/proof prose is present and non-trivial                                      | ✓      | Prose completeness score = 100.0%                            |
| Generated PDF compiles without fatal errors                                         | ✓      | L2 GR PDF exists at output/pdf/gr_branch_successor_standal   |
| Canonical cross-reference annotations are present                                   | ✓      | Canonical cross-reference annotations: 14/15 claims          |
| Equivalence certificate is reviewed                                                 | ✓      | This certificate constitutes the domain-expert review docume |
| Decision-log approval is recorded                                                   | ✓      | Decision-log approval recorded = True                        |

## Canonical Cross-Reference Map

| Machine ID         | Canonical        | Declared | Computed | Cross-check                              |
|--------------------|------------------|----------|----------|------------------------------------------|
| ob:GR.real         | ob:GR.real       | C        | C        | Discharged by thm:GR.realization         |
| thm:GR.realization | GR.Realization   | C        | C        | Corresponds to Stage 1 of the five-stage |
| ob:GR.deform       | ob:GR.deform     | C        | C        | Discharged by thm:GR.deformation         |
| thm:GR.deformation | GR.Deformation   | C        | C        | Stage 2 — deformation within the realize |
| ob:GR.compat       | ob:GR.compat     | C        | C        | Discharged by thm:GR.compat              |
| ob:GR.EFE1         | ob:GR.EFE1       | C        | C        | Discharged by thm:GR.compat (full discha |
| ob:GR.EFE2         | ob:GR.EFE2       | C        | C        | Discharged by thm:GR.compat (full discha |
| ob:GR.EFE3         | ob:GR.EFE3       | C        | C        | Discharged by thm:GR.compat (full discha |
| thm:GR.compat      | GR.Compatibility | C        | C        | Stage 3 — EFE compatibility and structur |

| Machine ID                  | Canonical      | Declared       | Computed       | Cross-check                                 |
|-----------------------------|----------------|----------------|----------------|---------------------------------------------|
| ob:GR.closure               | ob:GR.closure  | C              | C              | Discharged by thm:GR.domain at domain-bo    |
| thm:GR.domain               | GR.9.1         | C              | C              | Stage 4+5 — final domain-bounded closure    |
| thm:GR.CP.1                 | GR.CP.1        | C              | C              | Bundles realization+deformation+compat f    |
| thm:VI.9.1                  | —              | U              | U              |                                             |
| ob:GR.global-subcriticality | ob:GR.BF9      | 0              | 0              | ACTIVE FRONTIER — prevents promotion fro    |
| status:GR                   | prop:GR-status | domain-bounded | domain-bounded | achieved — generated analogue of the canoni |

## Five-Stage GR Closure Chain

The GR branch closes in five sequential stages, each discharging one obligation:

1. ob:GR.real discharged by thm:GR.realization [C]
2. ob:GR.deform discharged by thm:GR.deformation [C]
3. ob:GR.compat + ob:GR.EFE1/2/3 discharged by thm:GR.compat [C]
4. ob:GR.closure discharged by thm:GR.domain (= GR.9.1) [C]
5. Book VI legitimacy: thm:VI.9.1 [U] licenses the continuum transition

The global frontier ob:GR.global-subcriticality is *not* discharged. Domain-bounded CERT-CLOSE is achieved; global CERT-CLOSE is not claimed.

## Governance Boundary Conditions

- Domain-bounded CERT-CLOSE: **claimed** on the realized domain  $\mathcal{D}$
- Global CERT-CLOSE: **not claimed** (global curvature-subcriticality open)
- No hidden supplementation: all structure derived from declared spine imports
- Supersession effect: **none**; this is a release candidate, not a supersession